Solvency Field Theory

An Information-Theoretic Foundation for Cosmology and Gravity

Shaun Fosmark

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Abstract

The universe has a history because information processing is irreversible. Expansion is the shape of that history at cosmological scale, and gravity is its shape at every other scale.

Solvency Field Theory rests on one physical commitment: every particle in the universe maintains its quantum existence by processing information at a rate set by its energy, and the universe pays for that processing through geometry. A massive particle ticks at the Compton frequency mc^2/h. Each tick writes one irreversible bit to the holographic boundary. The local accumulation of ticks curves spacetime — that is gravity. The global sum of ticks forces the boundary to grow — that is expansion. There is no separate mechanism for each. Gravity and expansion are the same operation measured at different scales.

From this foundation, with no new particles, no new dimensions, no freely propagating dark sector, and no phenomenological fit parameters, the framework derives: a modified Friedmann equation $H^2 = \rho/(4\pi^2)$; a galactic acceleration threshold $a_{\mathrm{SFT}} = cH_0/(2\pi) = 1.04 \times 10^{-10}$ m/s^2 confirmed against 137 SPARC galaxies; a morphology-dependent interpolation coefficient supported at $p < 0.001$ across 131 galaxies; a dark-energy reconstruction matching DESI DR2; CMB compatibility at the calibrated baseline ($\Delta\chi^2 = 2 \times 10^{-6}$); a repaired single-field action whose static vacuum branch collapses exactly to Einstein–Hilbert and therefore recovers Schwarzschild; and a branch-level modified Tolman–Oppenheimer–Volkoff system for stellar interiors. The bookkeeping variable χ in the action is constrained and auxiliary, not a new propagating matter field.

This paper presents the theory in the order of physical inevitability rather than the order of historical discovery. The reader is first shown the mechanism at the single-particle level, then watches it scale up through two particles, many particles, and the continuum, arriving at the covariant source, the action, and the regime-by-regime consequences. The companion volume carries the full derivation archive.

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1 Introduction

The universe has a history because information processing is irreversible. Expansion is the shape of that history at cosmological scale, and gravity is its shape at every other scale.

This is the thesis of Solvency Field Theory, and this paper is a sustained argument for it.

The argument is not that gravity and expansion share a common mathematical origin or that they can be described by the same equations in certain limits. The argument is stronger: gravity and expansion are the same physical operation — the irreversible processing of information by every particle in the universe — viewed from two different scales. Locally, where the processing concentrates around massive objects, the accumulated cost curves the geometry. That curvature is gravity. Globally, where the processing is summed over everything that exists, the total cost forces the holographic boundary of the universe to grow. That growth is expansion.

If this is correct, then several problems in modern physics that are conventionally treated as separate mysteries dissolve into different measurements of a single quantity. The dark energy problem, the MOND acceleration scale in galaxies, and the holographic capacity gap are not three puzzles requiring three solutions. They are three regimes of one information-maintenance process.

Solvency Field Theory introduces no new particles, no new dimensions, no freely propagating dark sector, and no modifications to the gravitational dynamics of general relativity. The action contains one constrained auxiliary bookkeeping variable, but no new propagating matter field. It introduces one physical commitment: that existence is active, not passive, and that the cost of maintaining quantum state is paid through geometry. Everything else follows from applying this commitment across regimes and checking the results against data.

1.1 How this paper is organized

The paper presents the theory in the order of physical inevitability rather than the order of historical discovery. Sections 2 and 3 establish the motivation and the axiomatic foundation. Section 4 introduces the physical mechanism — the Compton Maintenance Postulate — at the single-particle level. Section 5 is the *mesoscale bridge*: the reader watches the mechanism scale from one particle to two particles to N particles to the continuum, seeing gravity emerge as a gradient and expansion as a sum. Section 6 derives the modified Friedmann equation, which by that point feels inevitable rather than postulated.

Sections 7 and 8 present the galactic confirmation and the structural identity $4\pi^2$. Section 9 presents the Lagrangian — the five-term covariant action that encodes the entire theory. Section 10 develops the forced thermodynamic transitions. Sections 11 and 12 deliver the late-universe and early-universe phenomenology. Section 13 presents the strong-field sector: exact Schwarzschild exterior recovery and the modified Tolman–Oppenheimer–Volkoff interior. Sections 14 and 15 state the predictions and the open theorems. Section 16 concludes.

The companion volume carries the full derivation chain for every result stated in this paper. The master paper carries the mechanism, the equations that define it, and the status of each claim. The companion carries the proofs.

2 Three Unsolved Problems

Three problems in modern cosmology are conventionally treated as separate puzzles. Solvency Field Theory proposes that they are three measurements of the same underlying quantity.

2.1 The dark energy problem

The universe is expanding at an accelerating rate. In the standard framework, this acceleration requires a component — dark energy — whose energy density is approximately 10^{-9} J/m^3. Quantum field theory predicts a vacuum energy density approximately 10^{120} times larger. This discrepancy between the predicted and observed values of the vacuum energy is the cosmological constant problem, and it is the largest quantitative disagreement in all of physics.

2.2 The MOND acceleration scale

Galactic rotation curves deviate from the predictions of Newtonian gravity at a characteristic acceleration scale $a_0 \approx 1.2 \times 10^{-10}$ m/s^2. This scale is empirically established across hundreds of galaxies spanning several orders of magnitude in mass, size, and morphology. It is not predicted by either general relativity or the standard dark matter paradigm. The Modified Newtonian Dynamics (MOND) framework treats a_0 as a fundamental constant of the theory, but does not derive it.

2.3 The holographic capacity gap

The holographic principle establishes that the maximum information content of a spatial region is proportional to its boundary area in Planck units, not its volume. The total information content of the observable universe — measured through the Bekenstein–Hawking entropy of its black holes and the thermodynamic entropy of its matter — appears to be a specific fraction of the boundary capacity. The nature of this fraction, and its connection to the expansion rate, has no explanation in the standard framework.

2.4 One identity, three regimes

Solvency Field Theory proposes that these three problems share a single geometric origin. The factor connecting the observed dark energy density to the matter content, the acceleration scale connecting galactic dynamics to the expansion rate, and the fraction connecting the information content to the boundary capacity are all expressions of one structural identity:

$$\boxed{4\pi^2} \tag{1}$$

This factor appears in five distinct physical contexts within the framework, and in each context decomposes into the same two independent factors of 2π for the same two physical reasons (Section 8). The three unsolved problems are not three mysteries. They are three windows onto the same conversion factor between quantum phase accounting and gravitational geometry.

3 The Foundation: Four Axioms and the Master Equation

Solvency Field Theory rests on four axioms. The first three are drawn from established physics. The fourth is the framework's closure axiom — the specific physical commitment that generates the theory.

3.1 Axiom 1: Information processing has an irreducible thermodynamic cost

This is Landauer's principle (1961). Erasing one bit of information dissipates at least $k_B T \ln 2$ of energy. Information processing is not free; it has a minimum thermodynamic cost that cannot be reduced by any physical process.

3.2 Axiom 2: Time is the macroscopic perception of irreversible state changes

This is the Connes–Rovelli thermal time hypothesis (1994). The direction and flow of time emerge from the thermodynamic arrow — from the irreversibility of state changes. Time is not a background parameter; it is what irreversible computation looks like from the inside.

3.3 Axiom 3: The holographic bound on information density is absolute

This is the holographic principle (Bekenstein 1973, 't Hooft 1993, Susskind 1995). The maximum information content of a spatial region is bounded by its boundary area measured in Planck units: $I_{\max} = A/(4\ell_P^2)$. No physical process can store more information in a region than its boundary can encode.

3.4 Axiom 4: Cosmic expansion is the geometric cost of maintaining an irreversible information ledger

This is the SFT closure axiom. It says: the universe processes information irreversibly (Axiom 1), this processing constitutes the flow of time (Axiom 2), the processed information must be stored on a boundary with finite capacity (Axiom 3), and as the total processed information grows, the boundary must grow to accommodate it. That boundary growth is cosmic expansion.

3.5 The master equation

From these four axioms, the master equation follows as a thermodynamic identity within the stated scaffold:

$$\boxed{H = \frac{\dot{S}}{I}} \tag{2}$$

The Hubble parameter H — the expansion rate of the universe — equals the ratio of the total information production rate $\dot{S}$ to the total boundary capacity I. This is not a postulated dynamical law. It is the statement that the rate of boundary growth equals the rate at which new information demands accommodation.

The master equation is the native thermodynamic identity of the framework. Everything that follows — the modified Friedmann equation, the galactic acceleration scale, the dark energy equation of state, the CMB compatibility, the covariant action, the strong-field closure — is a consequence of this identity applied in different regimes.

4 The Compton Maintenance Postulate

The master equation $H = \dot{S}/I$ requires a physical mechanism: what, concretely, is producing the information that forces the boundary to grow? The Compton Maintenance Postulate provides the answer.

> *Every massive particle maintains its quantum existence through continuous phase evolution at the Compton frequency mc^2/h. Each complete phase cycle constitutes one irreversible bit of information. Each bit demands one Planck area of holographic boundary surface to accommodate it.*

A particle does not simply *exist*. It *persists* — actively, continuously, and at a specific rate:

$$\boxed{\nu_C = \frac{mc^2}{h}} \tag{3}$$

A proton ticks at approximately 2.27×10^{23} Hz.[1] An electron ticks at approximately 1.24×10^{20} Hz. The proton is a louder writer by a factor of approximately 1836 — the mass ratio. A photon, which has no rest mass, does not tick in place; it hands off from cell to cell at the speed of light, spending its entire c-budget spatially rather than internally.

This postulate does not introduce a new interaction. It identifies a cost that was always present but never accounted for: the thermodynamic cost of maintaining quantum state, paid continuously, one bit per Compton cycle, one Planck area per bit.

A clarification is necessary here, because it addresses the first objection a physicist will raise. In ordinary quantum mechanics, phase evolution is unitary — it preserves information and is thermodynamically reversible. SFT does not claim that the bulk phase rotation itself is irreversible. The irreversible cost attaches to the boundary bookkeeping required to maintain persistent phase-locked state on a finite holographic ledger in an expanding universe. The Compton cycle supplies the count unit; the ledger update supplies the thermodynamic irreversibility. The distinction is between the unitary evolution of the quantum state (which is reversible) and the holographic registration of that evolution on a finite boundary (which is not).

The distinction between massive particles and photons is not a separate rule. It follows from the universal c-budget. Everything in the universe moves at c — the base processing rate, one Planck length per Planck time, one operation per tick. In SFT language, rest energy corresponds to internal phase cycling: a massive particle's c-limited processing budget is expressed partly or wholly as internal phase evolution rather than spatial propagation. For elementary particles, the Higgs mechanism contributes to rest mass; for composite particles such as protons and neutrons, binding and field energy contribute as well. SFT uses the total rest energy mc^2, not a Higgs-only origin of mass. The split between internal and spatial motion is what we call the mass–energy relation and what special relativity describes as time dilation.

A particle at rest uses its entire c-budget internally. It ticks at the full Compton rate and writes to the same local cell repeatedly. A moving particle splits its budget: some goes to internal ticking (at a reduced rate — time dilation), some goes to spatial transport. A photon uses its entire budget spatially, handing off from one cell to the next every Planck time.

The Compton Maintenance Postulate is the seed from which the entire framework grows. Everything that follows is the consequence of asking: what happens when many particles tick, when their costs accumulate, and when the geometry responds?

5 The Mesoscale Bridge: From One Particle to the Universe

The Compton Maintenance Postulate describes one particle. The modified Friedmann equation (Section 6) describes a universe full of fluid. Between these two levels — one particle and the cosmos — there is a gap. This section bridges it by counting. The argument proceeds from one particle to two particles to many particles to the continuum, and at each stage, no new physics enters. The macroscopic framework is the microscopic framework summed.

[1] Throughout this master paper, h denotes Planck's constant. In the companion volume, Planck's constant is written h_P to avoid confusion with the spatial projector $h^{\mu\nu}$.

5.1 One particle: existence as activity

A single massive particle at rest in otherwise empty space ticks at the Compton frequency $\nu_C = mc^2/h$. Its contribution to the total information processing rate of the universe is

$$\dot{s}_1 = \frac{mc^2}{h}. \tag{4}$$

The particle's local lag field ψ is the geometric footprint of this ticking, measured as the local departure from the cosmological background bookkeeping state. At the location of the particle, the maintenance cost per unit volume is concentrated. Moving outward, the cost decreases. In the weak-field limit, the local lag field reduces to the Green's-function solution of the Poisson equation for a point maintenance source. That solution is the standard $1/r$ gravitational potential, reinterpreted as the local maintenance-debt profile. The full derivation is carried in the companion volume.

There is nothing else here. No force, no field equation, no dynamics. Just a ticking particle and its geometric debt profile. Everything else emerges from counting.

5.2 Two particles: gravity as gradient

Place a second particle at distance r from the first. Each particle ticks at its own Compton frequency. Each generates its own local lag field. The total lag field at any point in space is the sum of the two individual lag fields:

$$\psi_{\text{total}}(\mathbf{x}) = \psi_1(\mathbf{x}) + \psi_2(\mathbf{x}). \tag{5}$$

This additivity is not an assumption. It is a consequence of the fact that information costs add — the total maintenance cost of two particles is the sum of their individual costs.

Now consider the gradient of the total lag field evaluated at the position of particle 1. The contribution from particle 1's own field is zero at its own location. The contribution from particle 2's field is

$$\nabla\psi_{\text{total}}\big|_{\text{particle 1}} = \nabla\psi_2\big|_{\text{particle 1}}, \tag{6}$$

which points from particle 1 toward particle 2.

This gradient is gravity. Not "produces gravity," not "sources gravity" — *is* gravity. Particle 1 sits on a slope in the maintenance-debt landscape created by particle 2's ticking. The slope tilts toward particle 2. In general relativity, we call this slope spacetime curvature. In SFT, we call it the gradient of accumulated maintenance debt. They are descriptions of the same geometric fact.

At this stage, Newton's law is already visible:

$$F = -\nabla\psi \propto \frac{m_1 m_2}{r^2}, \tag{7}$$

with the proportionality constant determined by the conversion between Compton ticking rate and geometric debt, which is where G enters.

5.3 N particles: bulk gravity and pressure

Continue to N particles:

$$\psi_{\text{total}}(\mathbf{x}) = \sum_{i=1}^{N} \psi_i(\mathbf{x}). \tag{8}$$

The gravitational field experienced by any particle j is the gradient of the sum of all other particles' lag fields:

$$\mathbf{g}_j = -\nabla\psi_{\text{total}}\big|_{\mathbf{x}_j} = -\sum_{i\neq j} \nabla\psi_i\big|_{\mathbf{x}_j}. \tag{9}$$

When particle j is surrounded by many other particles — in the interior of a star, a planet, a nucleus — the lag field contributions from all directions create a bowl. The gradient points inward. The deeper the bowl, the stronger the pull. This is bulk gravity, built from nothing more than overlapping debt profiles.

Now consider what happens when the particles are in motion. A moving particle splits its c-budget between internal ticking and spatial transport. The spatial component writes to successive cells rather than the same cell. When summed over many moving particles, this transport creates a momentum flux — and that flux is pressure.

Pressure is not a separate phenomenon from ticking. It is the spatial component of the ticking for particles that are in motion. A gas of particles bouncing around in a box has an energy density ρ (from the sum of all the ticking rates) and a pressure p (from the sum of all the transport rates). Both come from the same c-budget, split differently across the particle population.

This is why the total operation density is

$$\boxed{\rho_{\text{ops}} = \frac{\rho + 3p}{h}} \tag{10}$$

The ρ/h is the sum of all the ticking rates: every particle contributing mc^2/h, summed and coarse-grained into an energy density divided by h. The $3p/h$ is the sum of all the transport rates: every moving particle contributing its momentum-flux burden, summed over three spatial dimensions. There is one primitive operation — state-service at rate E/h — and two write patterns: same-cell refresh (mass) and next-cell handoff (transport). Their sum is the total source.

5.4 The continuum limit

Consider a coarse-graining volume $\mathcal{V}$ containing N particles with masses m_i and velocities v_i. The energy density is

$$\rho = \frac{1}{\mathcal{V}} \sum_{i\in\mathcal{V}} \gamma_i m_i c^2. \tag{11}$$

The pressure (for an isotropic velocity distribution) is

$$p = \frac{1}{3\mathcal{V}} \sum_{i\in\mathcal{V}} \gamma_i m_i v_i^2. \tag{12}$$

The operation density is

$$\rho_{\text{ops}} = \frac{1}{\mathcal{V}h} \sum_{i\in\mathcal{V}} \left(\gamma_i m_i c^2 + \gamma_i m_i v_i^2\right) = \frac{\rho + 3p}{h}, \tag{13}$$

where the first term in the parenthesis is the occupancy load (internal ticking) and the second is the transport burden (momentum flux), with the factor of 3 arising from the isotropic sum over spatial dimensions.

The transition from "N particles with lag fields" to "a continuous fluid with energy density ρ and pressure p" introduces no new physics. It is a coarse-graining that replaces a discrete sum with a smooth integral, valid whenever the particle count is large enough that statistical fluctuations are small. For stars, galaxies, and the cosmological fluid, this condition is always satisfied.

The covariant form of the same particle sum is

$$\rho_{\text{ops}} = \frac{T_{\mu\nu}u^\mu u^\nu + h^{\mu\nu}T_{\mu\nu}}{h}, \tag{14}$$

where u^μ is the clock congruence and $h^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$ is the spatial projector. The fluid-level description and the particle-level description are the same physics expressed at different scales.

5.5 Why the gradient is gravity and the sum is expansion

With the continuum limit established, the two macroscopic manifestations of Compton ticking can be stated precisely.

Gravity is the local gradient of the lag field. In a region where the energy density varies from point to point — near a star, near a galaxy — the lag field varies correspondingly. The gradient of the lag field is the gravitational field. This reduces to Poisson's equation in the Newtonian limit and to the full exponential metric in the strong-field limit. It is the continuum version of "each particle sits on the slope created by every other particle's ticking."

Expansion is the global sum of the lag field. The total information processing rate of the universe is the volume integral of ρ_{ops} over the Hubble volume. The holographic boundary must grow to accommodate this total. The rate of boundary growth is the expansion rate. The master equation $H = \dot{S}/I$ is the continuum version of "add up all the ticks and the boundary has to grow to fit them."

There is no third thing. No mechanism, no coupling, no bridge between gravity and expansion. They are the gradient and the integral of the same field. The transition between them — the regime where local structures can absorb their own maintenance costs versus forwarding them to the boundary — is governed by the acceleration scale $a_{\text{SFT}} = cH_0/(2\pi)$.

5.6 The mass defect and binding energy

The particle-level picture gives a natural interpretation of the mass defect. A bound system of N particles ticks slower than its free constituents would separately:

$$\dot{s}_{\text{bound}} = \frac{M_{\text{bound}}\, c^2}{h} = \sum_i \frac{m_i c^2}{h} - \frac{E_{\text{binding}}}{h}. \tag{15}$$

Binding reduces the maintenance cost. A bound system is cheaper to maintain than its free constituents because some internal phase structure is shared rather than independently serviced. The universe pays less to maintain a helium nucleus than two free protons and two free neutrons, and the difference is exactly the binding energy.

SFT does not derive the forces that produce binding — the strong force, electromagnetism, or the weak force remain the domain of the Standard Model. SFT's compatibility with the Standard Model is automatic: the lag field between two nucleons at femtometer separation produces a gravitational binding energy of order $Gm_p^2/r \sim 10^{-45}$ J, while the actual nuclear binding energy is of order 10^{-12} J. The lag field is 33 orders of magnitude too small to affect particle physics.

5.7 What this section establishes

The connecting rod between the Compton Maintenance Postulate and the cosmological master equation is counting. No new physics enters between one particle and a universe. The macroscopic framework is the microscopic framework summed.

6 The Modified Friedmann Equation

The reader has now watched the mechanism scale from one particle to the continuum, and has seen why the covariant operation density $\rho_{\text{ops}} = (\rho + 3p)/h$ is the natural source. The modified Friedmann equation follows.

6.1 From the master equation to the Friedmann ratio

In homogeneous FLRW cosmology, the master equation $H = \dot{S}/I$ can be evaluated explicitly. The total information production rate $\dot{S}$ is the integral of ρ_{ops} over the Hubble volume. The total boundary capacity I is determined by the holographic bound applied to the Hubble sphere. The ratio gives

$$\boxed{H^2 = \frac{\rho}{4\pi^2}} \tag{16}$$

in natural units where $8\pi G/3 = 1$. In standard units:

$$H^2 = \frac{8\pi G}{3} \cdot \frac{1}{4\pi^2} \cdot \rho. \tag{17}$$

This is the modified Friedmann equation of Solvency Field Theory. It differs from the standard Friedmann equation $H^2 = 8\pi G\rho/3$ by a single content-independent factor:

$$\frac{H^2_{\text{SFT}}}{H^2_{\text{std}}} = \frac{1}{4\pi^2} \approx 0.02533. \tag{18}$$

The factor $4\pi^2$ is not a fit parameter. It is derived from the axioms and the Compton Maintenance Postulate through the holographic count equation, which equates the total Compton phase cycles within a Hubble volume to the boundary area of that volume over one Hubble time. The derivation is in the companion volume.

6.2 The holographic count equation

The count equation is the specific form of the master equation applied to the Hubble horizon:

$$I(t_H) = S(t_H), \tag{19}$$

evaluated over one Hubble time $t_H = 1/H$. This says: the total information processed in one Hubble time equals the total boundary capacity. It is a settlement condition — the ledger balances over its natural timescale.

The $4\pi^2$ arises as the product of two independent factors of 2π:

$$4\pi^2 = (2\pi)_{\text{phase}} \times (2\pi)_{\text{thermal}}. \tag{20}$$

The first factor comes from how SFT counts information: one bit per complete Compton cycle, which is h of action, not $\hbar$. The ratio $h/\hbar = 2\pi$ enters the conversion. The second factor comes from

how horizons have temperature: the Unruh temperature carries a 2π through the KMS periodicity of quantum fields for accelerated observers. This two-factor decomposition is developed fully in Section 8.

6.3 Five formulations, one result

The modified Friedmann equation has been derived or recovered through five independent formulations:

Formulation	**Free parameters**	**Recovers $H^2 \propto \rho/(4\pi^2)$?**
Master equation (algebraic)	None	Yes
Holographic count equation	None	Yes (derived)
Constrained minisuperspace action	None	Yes (variational)
Single-field covariant action (dust)	None once count imposed	Yes (exact)
Single-field covariant action (radiation)	None once count imposed	Yes (asymptotically exact)

Each formulation reaches the same result from a different mathematical starting point. The agreement across five independent derivation paths is a consistency check: the modified Friedmann equation is not an artifact of a particular method.

6.4 What this section establishes

The modified Friedmann equation $H^2 = \rho/(4\pi^2)$ is derived with zero free parameters from the Compton Maintenance Postulate and the holographic count equation. It differs from the standard Friedmann equation by a single geometric factor that decomposes into two independent physical origins. The equation has been derived or recovered through five independent formulations.

End of Part 1 — Sections 1 through 6. Part 2 continues with the galactic confirmation (Section 7), the $4\pi^2$ structural identity (Section 8), and the Lagrangian (Section 9).

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